

SAGE: Self-Evolving Acquisition through Guided Escalation for Subtle Anomaly Segmentation

Anonymous submission

Abstract

Detecting subtle, weak-signal structures in scientific imaging, such as low-contrast irregularities in CT scans, faint abnormalities in microscopy, or rare targets in large survey data, is often limited less by model capacity than by the cost of expert annotation. The structures of interest are small, weakly visible, and frequently predicted as background, so standard active learning methods based on uncertainty or diversity can overlook precisely the regions most useful for improving the next model. We present **SAGE**, a self-evolving active learning framework for region-level subtle anomaly segmentation under sparse supervision. SAGE conditions acquisition on the current segmentation state, uses a fast vision-language policy to rank candidate regions, and selectively escalates ambiguous or redundancy-sensitive decisions to a fixed deliberative planner. After annotation and retraining, reward-set improvements are converted into pairwise preferences that update the fast policy through outcome-validated preference learning. On CT-Defect and two public subtle-defect benchmarks, MVTEC AD and VisA, SAGE improves subtle-defect Dice by 6.6–7.2 points over the strongest learned region-selection baseline and surfaces confirmed targets earlier under a fixed review budget. These results show that acquisition policies can be improved from measured downstream segmentation outcomes, providing a data-efficient mechanism for focusing scarce expert supervision on subtle visual evidence. Anonymous reproducibility material, including split manifests, prompt templates, and extended implementation details, is provided at <http://anonymous.4open.science/r/SAGE-Self-Evolving-Acquisition-through-Guided-Escalation-for-Subtle-Anomaly-Segmentation-9504/>.

Introduction

Many scientific imaging problems require detecting subtle, weak-signal structures from limited expert supervision (Wang et al. 2023). Examples include low-contrast structures in CT and tomography scans, faint cellular abnormalities in microscopy, small structural defects in inspection imagery, localized features in environmental or physical-system imagery, and rare targets in large-scale scientific image surveys (Litjens et al. 2017; Moen et al. 2019; Bergmann et al. 2019; Zou et al. 2022; Willard et al. 2022; Lochner and Bassett 2021). In these settings, the structures of interest are often small, spatially localized, visually similar to background, and expensive to annotate. The bottle-

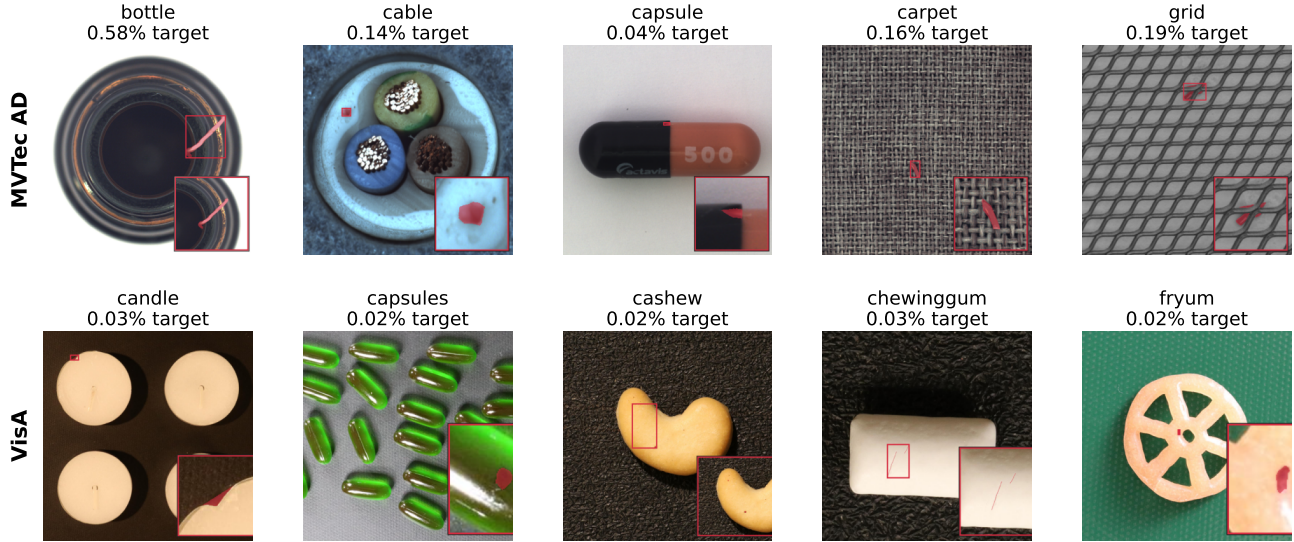
neck is therefore not only model capacity, but also deciding which regions should receive scarce expert labels so that each annotation produces maximal downstream utility (Settles 2009; Gal, Islam, and Ghahramani 2017; Sener and Savarese 2017).

Subtle anomaly segmentation is a concrete instance of this broader scientific-imaging challenge. As illustrated in Figure 1, the relevant structures can occupy only a small spatial region while remaining visually similar to surrounding background. Unlike conventional segmentation targets, subtle anomalies occupy only a small fraction of the image, are often low-contrast or irregular, and can be visually similar to normal background structure (Bergmann et al. 2019; Zou et al. 2022). Reliable pixel-level annotation is expert-intensive, making label efficiency a central requirement when adapting segmentation models to new domains, sensors, or experimental conditions (Settles 2009; Lüth et al. 2023). This is especially important in scientific and engineering settings where additional labels may require expert review, specialized instruments, or repeated experimental campaigns (Karniadakis et al. 2021; Willard et al. 2022).

Active learning (AL) addresses this problem by selecting a small informative subset of the unlabeled pool for annotation (Settles 2009). Most AL methods rely on fixed acquisition proxies such as predictive uncertainty (Gal, Islam, and Ghahramani 2017), feature-space diversity or coverage (Sener and Savarese 2017; Yehuda et al. 2022), or combinations of uncertainty and representativeness. These criteria implicitly assume that uncertain or diverse samples are also the samples that most improve the next model. This assumption is fragile for subtle structures: a target may occupy only a few pixels among a large background, so image-level uncertainty can remain low even when the target pixels are confidently mislabeled. Diversity scores can likewise favor visually unusual regions that do not repair the model’s current failure modes (Lüth et al. 2023; Melo et al. 2024).

The core issue is that annotation value is not an intrinsic property of an image or region; it depends on the current model state. Two visually similar regions may have different value depending on the model’s predictions, uncertainty, dominant failure mode, and previous annotation history. Although selecting by expected model effect is a classical idea (Roy and McCallum 2001), and recent work shows that annotation value can shift across target models

Subtle visual targets under sparse expert supervision



Red overlay marks the target mask; inset shows a zoomed view of the localized structure.

Figure 1: Examples of subtle target regions from MVTec AD and VisA (Bergmann et al. 2019; Zou et al. 2022). Red overlays show available ground-truth masks, and insets show zoomed views of the localized structures. The targets occupy only a small fraction of the image, which illustrates why image-level uncertainty or diversity can miss regions that are important for improving the next segmentation model.

and training states (Tang and Huang 2022), many practical AL systems still rely on static proxies. Learned acquisition methods address this limitation by regressing error reduction, learning selection policies, or choosing among AL algorithms (Konyushkova, Sznitman, and Fua 2017; Fang, Li, and Cohn 2017; Zhang et al. 2023), including for segmentation (Casanova et al. 2020). However, these approaches typically rely on proxy rewards, engineered state features, or task-agnostic signals, and do not explicitly learn acquisition behavior from measured post-retraining segmentation outcomes.

We present **SAGE**, a self-evolving active learning framework for region-level subtle anomaly segmentation under sparse expert supervision. SAGE makes three design choices for this regime. First, it acquires annotations at the region level so labeling effort is concentrated on spatially localized evidence rather than entire images. Second, it conditions acquisition on a round-wise model state containing current predictions, predictive uncertainty, failure modes estimated from labeled feedback, and prior annotation experience. Third, it separates fast local scoring from global batch construction: a fine-tunable vision-language policy ranks candidate regions, while a fixed deliberative planner is invoked only when redundancy, failure coverage, or ambiguity requires interaction-aware reasoning. After annotation and retraining, measured improvements on a held-out reward set are converted into pairwise preferences that update the fast policy through outcome-validated preference learning.

This use of preference learning differs from active learning for preference optimization, where the goal is to choose

human preference pairs for aligning a downstream language model (Muldrew et al. 2024; Kveton et al. 2025; Melo et al. 2024). In SAGE, preference optimization trains the acquisition policy itself: preferences are automatically derived from post-retraining segmentation outcomes. Technically, we use a Bradley-Terry pairwise ranking objective (Bradley and Terry 1952) parameterized by a DPO-style implicit reward (Rafailov et al. 2023), avoiding direct comparison of raw utility magnitudes across active-learning rounds. This allows the acquisition policy to evolve from measured downstream utility rather than from a fixed uncertainty, diversity, or planner-imitation signal.

Our contributions are threefold. First, we formulate subtle scientific visual segmentation as region-level, model-state-conditioned active learning, targeting the sparse-supervision regime where full-image annotation is costly and small structures are easily missed. Second, we introduce a self-evolving acquisition framework that combines a learned fast policy, selective deliberative planning, and outcome-validated preference learning from measured segmentation improvement. Third, we evaluate SAGE on MVTec AD, VisA, and CT-Defect, showing improved label efficiency and 6.6–7.2 point gains in subtle-defect Dice over the strongest learned region-selection baseline across the three datasets. On the two public benchmarks, the gains are 7.1–7.2 points, with additional analysis showing non-uniform category-level improvements and failure cases where gains shrink when static baselines already surface salient regions.

Related Work

Active learning and learned acquisition. Pool-based active learning commonly selects samples using fixed informativeness or representativeness proxies, including predictive uncertainty (Settles 2009; Gal, Islam, and Ghahramani 2017), core-set coverage (Sener and Savarese 2017), and feature-space diversity (Yehuda et al. 2022). Effect-based methods instead query samples expected to reduce model error (Roy and McCallum 2001), while learned acquisition methods regress error reduction, learn selection policies, or select among active-learning algorithms (Konyushkova, Sznitman, and Fua 2017; Fang, Li, and Cohn 2017; Zhang et al. 2023). SAGE builds on this line of work but differs in the feedback signal: the acquisition policy is conditioned on the current segmentation state and updated from measured post-retraining segmentation outcomes rather than from static uncertainty, diversity, or proxy reward signals.

Active learning for segmentation. Segmentation active learning differs by annotation granularity. Image-level methods select full images using uncertainty, diversity, adversarial representation learning (Sinha, Ebrahimi, and Darrell 2019), or ensemble disagreement (Yang et al. 2017), while region- and pixel-level methods query localized supervision using viewpoint uncertainty, difficulty estimates, or sparse pixel labels (Xie et al. 2020; Shin, Xie, and Albanie 2021). RALIS is closest to our setting because it learns a region-selection policy for segmentation (Casanova et al. 2020). However, RALIS uses a proxy reward and a local policy, whereas SAGE learns from realized downstream improvement and uses selective deliberative planning to construct redundancy-aware annotation batches. This distinction is important for subtle targets, which are small, low-contrast, and often confidently missed by uncertainty-driven selection.

Sparse scientific imaging, preference learning, and planning. Label scarcity is central in scientific imaging, including medical, microscopy, physical-system, and astronomical anomaly-detection settings (Litjens et al. 2017; Moen et al. 2019; Willard et al. 2022; Lochner and Bassett 2021). These domains share SAGE’s motivating regime: rare, weak-signal targets under expensive expert supervision. Our use of preference learning is distinct from active preference labeling for language-model alignment (Muldrew et al. 2024; Kveton et al. 2025; Melo et al. 2024). Instead, SAGE derives pairwise preferences automatically from segmentation outcomes and optimizes a Bradley–Terry/DPO-style objective (Bradley and Terry 1952; Rafailov et al. 2023). The planner is related to budgeted and submodular batch selection (Mao et al. 2024; Sener and Savarese 2017), but operates over learned region values and is invoked only for escalated decisions. Finally, although MVTec AD and VisA are often used for unsupervised anomaly detection (Bergmann et al. 2019; Zou et al. 2022), we study label-efficient supervised segmentation, where acquisition quality is the primary bottleneck.

SAGE: Outcome-Validated Region Acquisition

Problem Setup

We consider pool-based active learning for subtle scientific visual segmentation with *region-level* annotation. Each image is partitioned into a fixed grid of candidate regions, and labeling a region reveals pixel-level labels only inside that region. This matches settings where the structures of interest are spatially localized and expert annotation is costly. At round t , the acquisition pool contains labeled regions $\mathcal{L}_t$ and unlabeled regions $\mathcal{U}_t$. Starting from an initial seed $\mathcal{L}_0$, the learner queries b regions per round for T rounds. The segmentation model at round t is

$$M_t = f(\mathcal{L}_t),$$

where the training loss is computed only over labeled regions.

SAGE separates the data used for acquisition feedback from the data used for model selection and final reporting. A fixed labeled reward set $\mathcal{R}$ provides post-retraining feedback for acquisition-policy updates; a separate validation set $\mathcal{V}$ is used only for hyperparameter selection and early stopping; and the test set $\mathcal{T}$ is used only for final evaluation. The ideal round-wise objective is to select a batch whose annotation most improves the next model on $\mathcal{R}$:

$$\mathcal{B}_t^* = \arg \max_{\mathcal{B} \subseteq \mathcal{U}_t, |\mathcal{B}|=b} \mathcal{P}_{\mathcal{R}}(f(\mathcal{L}_t \cup \mathcal{B})),$$

where $\mathcal{P}_{\mathcal{R}}$ denotes segmentation quality on the reward set. This objective is not directly optimized or used for final reporting; it defines the outcome signal used to train the acquisition policy. The goal is therefore not to maximize uncertainty or diversity directly, but to learn which annotations produce measurable downstream segmentation improvement.

Framework Overview

SAGE is a round-wise acquisition framework with a learned fast policy and a fixed deliberative planner. The controller first summarizes the current segmentation state as

$$\mathcal{S}_t = \{P_t, H_t, F_t, E_t\}, \quad (1)$$

where P_t is the current prediction, H_t is predictive uncertainty, F_t is the dominant failure mode estimated only from labeled feedback data, and E_t is an experience memory of previously annotated regions and their observed utilities. No ground-truth labels from $\mathcal{U}_t$ are used to construct $\mathcal{S}_t$.

Figure 2 summarizes the full round-wise loop. The top path performs acquisition: the current model state is used to score unlabeled regions, construct a candidate shortlist, and compose the final annotation batch. The bottom path performs policy evolution: after annotation and retraining, reward-set improvement is converted into region utilities and within-round preferences that update only the fast policy.

Given a candidate region $r \in \mathcal{U}_t$, the fast vision–language policy receives the region crop, local context, and $\mathcal{S}_t$, and generates a structured response containing a human-readable rationale and an estimated annotation value $\hat{v}_t(r) \in [0, 1]$. The top-ranked regions form a shortlist

$$\mathcal{C}_t = \text{TopK}(\hat{v}_t(r)),$$

SAGE: outcome-validated region acquisition

A learned fast policy selects regions; selective planning forms batches; reward-set improvement updates the policy.

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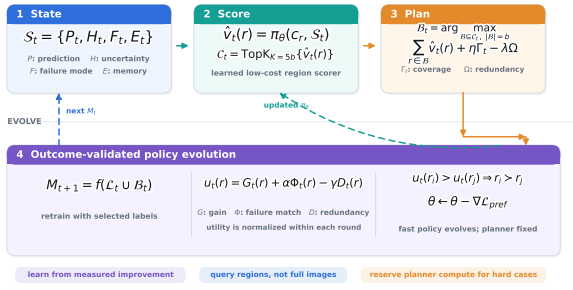


Figure 2: SAGE acquisition loop. The current model defines a compact state S_t , the learned fast policy scores candidate regions, selective planning constructs the annotation batch, and reward-set improvement induces preferences that update the fast policy. Solid arrows show the acquisition path; dashed arrows show the feedback path for the next round.

with $K = 5b$. Confident, non-redundant regions are accepted directly; ambiguous regions, near ties (within an escalation margin δ , specified in the appendix), and regions poorly represented in the experience memory are escalated to the planner. This separates low-cost local scoring from higher-cost batch-level reasoning.

Selective Planning

The planner composes the final annotation batch from the shortlist by balancing predicted value, failure coverage, and redundancy:

$$\mathcal{B}_t = \arg \max_{\mathcal{B} \subseteq \mathcal{C}_t, |\mathcal{B}|=b} \left[\sum_{r \in \mathcal{B}} \hat{v}_t(r) + \eta \Gamma_t(\mathcal{B}) - \lambda \Omega(\mathcal{B}) \right]. \quad (2)$$

Here Γ_t rewards coverage of the dominant failure mode F_t , and Ω penalizes semantically or spatially redundant regions. We solve Eq. (2) greedily, starting from directly accepted regions and adding the candidate with largest marginal gain until the budget is filled. The planner remains fixed throughout training; only the fast policy evolves. This design isolates the contribution of outcome-validated policy learning while keeping the expensive deliberative component stable and bounded.

Outcome-Validated Policy Evolution

After $\mathcal{B}_t$ is annotated, the segmentation model is retrained once to obtain $M_{t+1} = f(\mathcal{L}_t \cup \mathcal{B}_t)$. SAGE then measures improvement on the reward set and assigns each selected region an observed utility

$$u_t(r) = G_t(r) + \alpha \Phi_t(r) - \gamma D_t(r), \quad (3)$$

where $G_t(r)$ is outcome-conditioned credit, transferred by feature similarity from reward-set regions whose loss decreased after retraining; $\Phi_t(r)$ measures match to the dominant failure mode; and $D_t(r)$ penalizes redundancy with

same-round selections and the experience memory. Full definitions of G_t , Φ_t , and D_t are given in the appendix.

The observed utilities induce within-round pairwise preferences:

$$r_i \succ r_j \iff u_t(r_i) > u_t(r_j).$$

We use relative preferences rather than raw utility magnitudes because absolute improvements shrink and shift as the segmentation model improves. For a validated pair $r_i \succ r_j$, the response generated for r_i is treated as preferred over the response generated for r_j . The fast policy is then updated with a Bradley–Terry pairwise ranking objective parameterized by a DPO-style implicit reward. Thus, SAGE learns from annotation decisions that are verified by downstream segmentation improvement rather than from static uncertainty, diversity, or direct planner imitation.

In this work the deliberative planner is fixed and credit is assigned by outcome-conditioned similarity. This keeps the method lightweight and isolates whether the fast acquisition policy can evolve from measured segmentation outcomes; we discuss the resulting limitations in the supplementary.

Experimental Setup

Datasets. We evaluate SAGE on three subtle visual segmentation settings. MVTec AD contains 5,354 industrial inspection images from 15 categories with pixel-level masks (Bergmann et al. 2019), and VisA contains 10,821 images from 12 industrial categories (Zou et al. 2022). We use both as controlled testbeds for small, low-contrast, spatially localized targets under sparse annotation. Since their official splits are designed for unsupervised anomaly detection, we construct a supervised active-learning protocol by pooling images and forming category- and anomaly-stratified splits: 50% active-learning pool, 15% reward set $\mathcal{R}$, 15% validation set $\mathcal{V}$, and 20% held-out test set $\mathcal{T}$. These are supervised active-learning comparisons, not standard MVTec/VisA unsupervised leaderboard results. CT-Defect is a proprietary CT defect-segmentation dataset with approximately 13,562 grayscale tomography scans containing weak-contrast inclusion, void, and crack defects. We retain its official train/test split and carve reward and validation sets from the training split.

Protocol. All methods use the same SegFormer-B2 backbone, region grid, initial labels, annotation budget, retraining schedule, and five matched active-learning seeds. Images are resized to 512×512 and split into overlapping 128×128 candidate regions with stride 64. We start from 5% labeled regions and acquire 5% more per round for $T = 9$ rounds, yielding a final active labeled fraction of 50%. The reward set is used only to construct outcome feedback; validation is used only for hyperparameter selection and early stopping; and the test set is used only for final evaluation. Subtle defects are connected defect components whose area is below 1% of the image.

Baselines and metrics. We compare against Random, Entropy, BALD (Houlsby et al. 2011), CoreSet (Sener and Savarese 2017), VAAL (Sinha, Ebrahimi, and Darrell 2019), EquAL (Golestaneh and Kitani 2020), and RALIS (Casanova et al. 2020). We also include Frozen

VLM, which uses the same fast policy without outcome-validated updates, and Planner-All, which sends all short-listed candidates to the deliberative planner. We report Dice, IoU, AULC, subtle-defect Dice, subtle AULC, Review@K, Regions-to-90%, and defect-area error. Full hyperparameters, prompt format, reward-set accounting, and compute details are provided in the appendix.

Results

All results are reported as mean $\pm$ standard deviation over five matched active-learning seeds. The active annotation budget counts queried training regions; reward-set calibration, credit validation, operational costs, per-category results, and additional ablations are reported in the supplementary.

Overall label efficiency. Table 1 reports final segmentation performance. SAGE improves over uncertainty-, diversity-, adversarial-, and learned-policy baselines on all three datasets. RALIS is the strongest baseline overall, but SAGE improves Dice by 3.2 points on MVTec AD, 3.4 points on VisA, and 2.9 points on CT-Defect. AULC follows the same general trend, indicating that SAGE improves the utility of acquired labels across the active-learning process rather than only at the final round.

Subtle structures. Table 2 evaluates the target regime: connected target components occupying less than 1% of the image area. SAGE gives larger gains on these small structures than on overall Dice, improving subtle-region Dice over RALIS by 7.1 points on MVTec AD, 7.2 points on VisA, and 6.6 points on CT-Defect. The gains are not uniform across categories or rounds, but the average improvement is largest in the small-target regime. This supports the claim that model-state-conditioned region acquisition is most useful when relevant structures are weak, localized, and easily missed by uncertainty-driven acquisition.

Component analysis. Table 3 isolates the main design choices. Removing the planner lowers Dice and subtle Dice, showing that local scores alone do not reliably compose the best annotation batch. Frozen VLM performs better than static baselines but worse than SAGE, indicating that the gains are not simply from using a VLM as a fixed heuristic. Planner-All is slightly more accurate than SAGE but requires substantially higher cost, supporting selective escalation as the better accuracy–cost trade-off. Region-level acquisition is also essential: replacing it with image-level acquisition causes the largest drop, especially on subtle structures, confirming that spatially localized labeling is what concentrates supervision on small targets.

Learning dynamics and failure cases. Figure 3 shows Dice over acquisition rounds on MVTec AD and VisA. All methods share the same initial labeled regions, so round-0 performance is identical before acquisition begins. SAGE improves earlier and remains above the strongest baselines over most of the budget range, but the curves are not monotonic: small drops occur under repeated retraining with limited partial supervision. SAGE also does not help equally in every case. Its smallest gains occur when targets are large, high-contrast, or already surfaced early by uncertainty-based acquisition, and when the reward set contains few close ana-

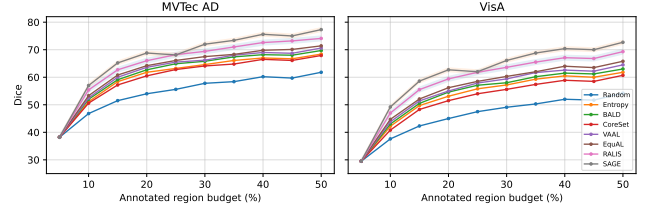


Figure 3: Active-learning curves on MVTec AD and VisA. Curves show mean Dice over five matched seeds; shaded bands denote standard deviation for SAGE and RALIS. Small non-monotonic fluctuations are expected under repeated retraining with limited partial supervision.

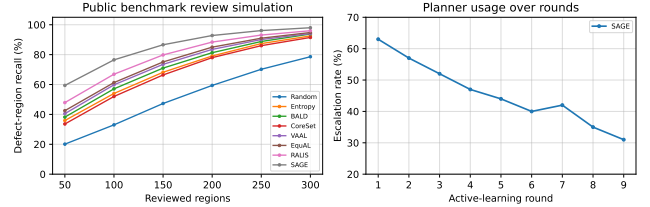


Figure 4: Offline review-prioritization simulation on public benchmarks. Left: target-region recall as a function of reviewed regions, averaged over MVTec AD and VisA. Right: planner escalation rate across active-learning rounds. SAGE surfaces confirmed target regions earlier while requiring fewer planner calls as the fast acquisition policy improves.

logues of the current failure mode. Per-category results and qualitative examples are provided in the supplementary.

Acquisition compute. Because all methods use the same SegFormer-B2 retraining schedule, the main compute difference comes from acquisition scoring and planning. Table 4 compares the VLM-based variants. SAGE uses substantially less acquisition compute than Planner-All because it escalates only ambiguous or redundancy-sensitive decisions to the larger planner. Compared with Planner-All, SAGE reduces relative VLM cost from 1.00 to 0.54 and selection time from 27.8 to 14.9 minutes per round, while retaining nearly the same segmentation accuracy (Table 3). Frozen VLM is cheaper than Planner-All but also less accurate, showing that the learned fast policy and selective planner are both needed for a favorable accuracy–compute trade-off.

Resource-Aware Scientific Discovery

SAGE is a review-prioritization layer, not an autonomous decision system. It allocates scarce expert attention to candidate regions most likely to improve the next segmentation model, matching scientific imaging settings where evidence is sparse, localized, and costly to annotate. The goal is to help experts inspect the most useful regions first, not to remove expert judgment from the workflow.

SAGE uses a fixed reward set for outcome feedback. This set can come from archived expert labels, adjudicated cases, or a small calibration study; it is not used to train the segmentation model. Instead, it measures whether newly ac-

Table 1: Main active-learning results at the final annotation budget. AULC summarizes Dice across acquisition rounds. Values are mean $\pm$ std over five matched seeds.

Method	MVTec AD		VisA		CT-Defect	
	Dice	AULC	Dice	AULC	Dice	AULC
Random	61.8 $\pm$ 1.8	53.4 $\pm$ 1.9	54.2 $\pm$ 2.0	46.7 $\pm$ 1.9	58.6 $\pm$ 1.9	49.8 $\pm$ 2.0
Entropy	68.4 $\pm$ 1.3	59.9 $\pm$ 1.5	61.8 $\pm$ 1.6	53.4 $\pm$ 1.6	65.9 $\pm$ 1.5	57.4 $\pm$ 1.5
BALD	69.7 $\pm$ 1.4	61.2 $\pm$ 1.5	63.0 $\pm$ 1.5	54.6 $\pm$ 1.5	67.2 $\pm$ 1.4	58.8 $\pm$ 1.5
CoreSet	67.9 $\pm$ 1.5	60.4 $\pm$ 1.4	60.7 $\pm$ 1.7	52.8 $\pm$ 1.7	64.8 $\pm$ 1.6	56.7 $\pm$ 1.7
VAAL	70.6 $\pm$ 1.2	62.5 $\pm$ 1.3	64.5 $\pm$ 1.4	56.2 $\pm$ 1.4	68.1 $\pm$ 1.3	60.1 $\pm$ 1.4
EquAL	71.4 $\pm$ 1.1	63.7 $\pm$ 1.2	65.8 $\pm$ 1.3	57.1 $\pm$ 1.3	69.0 $\pm$ 1.2	61.0 $\pm$ 1.3
RALIS	74.1 $\pm$ 1.0	66.1 $\pm$ 1.1	69.3 $\pm$ 1.1	60.9 $\pm$ 1.1	72.5 $\pm$ 1.0	63.9 $\pm$ 1.0
SAGE	77.3 $\pm$ 0.9	68.9 $\pm$ 0.9	72.7 $\pm$ 1.0	64.2 $\pm$ 1.0	75.4 $\pm$ 0.9	66.7 $\pm$ 1.0

Table 2: Subtle-region results. Subtle Dice is computed only over connected target components whose area is below 1% of the image. Values are mean $\pm$ std over five matched seeds.

Method	MVTec AD		VisA		CT-Defect	
	Subtle Dice	Subtle AULC	Subtle Dice	Subtle AULC	Subtle Dice	Subtle AULC
Random	38.9 $\pm$ 2.3	31.0 $\pm$ 2.1	31.6 $\pm$ 2.6	24.7 $\pm$ 2.4	35.4 $\pm$ 2.5	27.9 $\pm$ 2.3
Entropy	47.6 $\pm$ 2.0	40.2 $\pm$ 1.9	40.1 $\pm$ 2.2	33.4 $\pm$ 2.0	43.8 $\pm$ 2.1	36.5 $\pm$ 2.0
BALD	49.1 $\pm$ 1.9	41.5 $\pm$ 1.8	41.7 $\pm$ 2.1	35.0 $\pm$ 1.9	45.9 $\pm$ 2.0	38.1 $\pm$ 1.9
CoreSet	46.8 $\pm$ 2.1	39.1 $\pm$ 2.0	38.5 $\pm$ 2.4	31.9 $\pm$ 2.2	41.6 $\pm$ 2.3	34.1 $\pm$ 2.1
VAAL	51.3 $\pm$ 1.7	43.7 $\pm$ 1.7	44.0 $\pm$ 1.9	37.2 $\pm$ 1.8	47.7 $\pm$ 1.9	39.8 $\pm$ 1.8
EquAL	52.6 $\pm$ 1.6	45.0 $\pm$ 1.6	45.5 $\pm$ 1.8	38.5 $\pm$ 1.7	48.9 $\pm$ 1.8	41.0 $\pm$ 1.7
RALIS	56.4 $\pm$ 1.3	49.2 $\pm$ 1.4	50.6 $\pm$ 1.5	43.0 $\pm$ 1.5	53.7 $\pm$ 1.4	45.8 $\pm$ 1.4
SAGE	63.5 $\pm$ 1.2	55.7 $\pm$ 1.2	57.8 $\pm$ 1.3	50.1 $\pm$ 1.2	60.3 $\pm$ 1.2	52.0 $\pm$ 1.3

Table 3: Ablation study averaged over MVTec AD and VisA. Relative VLM cost is normalized so that Planner-All equals 1.0.

Variant	Dice	AULC	Subtle Dice	Cost $\downarrow$
SAGE	75.0 $\pm$ 0.8	66.6 $\pm$ 0.9	60.7 $\pm$ 1.1	0.54
w/o planner	72.4 $\pm$ 1.1	63.9 $\pm$ 1.2	55.0 $\pm$ 1.4	0.29
Planner-All	75.5 $\pm$ 0.8	66.9 $\pm$ 0.9	61.3 $\pm$ 1.1	1.00
Frozen VLM	72.9 $\pm$ 1.1	64.1 $\pm$ 1.2	54.8 $\pm$ 1.5	0.72
Image-level acquisition	69.5 $\pm$ 1.4	60.7 $\pm$ 1.5	45.9 $\pm$ 1.9	0.48

Table 4: Acquisition compute averaged over MVTec AD and VisA. Selection time excludes SegFormer retraining, which is identical across methods. Relative VLM cost is normalized so that Planner-All equals 1.0.

Method	Escalation	Rel. Cost $\downarrow$	Select Time $\downarrow$
Frozen VLM	51.8 $\pm$ 6.4%	0.72 $\pm$ 0.05	19.7 $\pm$ 1.5 min
Planner-All	100.0 $\pm$ 0.0%	1.00 $\pm$ 0.00	27.8 $\pm$ 1.9 min
SAGE	43.6 $\pm$ 5.8%	0.54 $\pm$ 0.04	14.9 $\pm$ 1.1 min

quired regions improve downstream performance. Reward-set sensitivity and calibration cost are reported in the supplementary material.

Figure 4 evaluates whether SAGE surfaces confirmed target-positive regions earlier under a fixed review budget. Candidate regions are ranked as if presented to an expert, and public masks are used retrospectively to measure target recall. SAGE recovers relevant regions earlier than uncertainty-, diversity-, adversarial-, and learned region-selection baselines, while planner escalation decreases across rounds. This supports the claim that outcome-validated acquisition can concentrate limited supervision on regions with higher downstream utility.

Conclusion

We presented SAGE, a self-evolving active learning framework for region-level subtle anomaly segmentation in scien-

tific imaging. SAGE addresses a key failure mode of conventional active learning: small, low-contrast targets can be confidently predicted as background and missed by uncertainty- or image-level selection. By conditioning acquisition on model state, combining a learned fast policy with selective planning, and updating from outcome-validated preferences, SAGE learns from annotation decisions verified by downstream improvement. Across MVTec AD, VisA, and CT-Defect, SAGE improves label efficiency and gives its largest gains in the subtle-target regime, with 6.6–7.2 point subtle-Dice improvements over the strongest learned region-selection baseline. Together with lower acquisition cost and earlier target surfacing, these results support outcome-validated acquisition as a resource-aware direction for scientific discovery under sparse expert supervision.

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Supplementary Material

Reproducibility

Code, split manifests, and prompt templates will be released publicly on acceptance. This section documents the experimental protocol, implementation details, and hyperparameters used to reproduce the reported results.

Dataset Splits

For MVTEC AD and VisA, the official splits are designed for unsupervised anomaly detection and place anomalous images only in the official test split. Since SAGE studies supervised region-level active learning, we construct category- and anomaly-stratified train/reward/validation/test splits. We use 50% of images for the active-learning training pool, 15% for the reward set $\mathcal{R}$, 15% for validation $\mathcal{V}$, and 20% for the held-out test set $\mathcal{T}$. Normal images are treated as all-background masks and are split using the same stratification. All methods share the same split for each seed.

For CT-Defect, we retain the official 10,453/3,109 train/test split. Reward and validation sets are carved from the training split, and the official test split is used only for final reporting.

Active-Learning Protocol

All methods use the same active-learning schedule, region grid, segmentation backbone, initial labeled regions, random seeds, and retraining protocol. Images are resized to 512×512 and partitioned into overlapping 128×128 candidate regions with stride 64, yielding at most 49 candidate regions per image. Region annotations reveal labels only inside the selected region, and the segmentation loss is masked outside labeled regions.

Segmentation Model and Training

The segmentation backbone is SegFormer-B2 with an ImageNet-pretrained MiT-B2 encoder. The segmentation model is retrained from the same initialization at each active-learning round using the currently labeled regions. The same training schedule is used for all acquisition methods.

Policy and Planner Hyperparameters

The fast acquisition policy is Qwen2.5-VL-3B-Instruct. The deliberative planner is frozen Qwen2.5-VL-7B-Instruct. The planner is not fine-tuned; only the fast acquisition policy is updated through outcome-validated preference learning.

VLM Input and Output Format

Each candidate region is presented to the fast policy as an image crop with surrounding context, the current segmentation prediction P_t , the uncertainty summary H_t , the dominant failure descriptor F_t , and retrieved examples from the experience memory E_t . The VLM output is parsed from a constrained JSON-style response:

```
{
  "rationale": short text,
  "value": [0, 1],
  "failure_match": [0, 1]
}
```

Malformed outputs are repaired by deterministic parsing when possible. If parsing fails, the candidate is assigned the lowest valid priority and is not escalated unless selected by another criterion. Experience examples are retrieved from E_t by cosine similarity in descriptor space, using the top three non-redundant prior regions.

Additional Method Details

Round-Wise Model State

At the start of round t , the active-learning controller constructs

$$\mathcal{S}_t = \{P_t, H_t, F_t, E_t\},$$

where P_t denotes current segmentation predictions, H_t denotes predictive uncertainty, F_t denotes the dominant failure mode estimated from labeled feedback data, and E_t denotes the experience memory.

The dominant failure mode F_t is computed only from regions whose labels are available: the reward set $\mathcal{R}$ and historical labeled regions. We compare M_t 's predictions with available masks, extract descriptors from high-loss and low-confidence regions, cluster those descriptors, and use the most prevalent cluster as the current dominant failure mode. No ground-truth labels from the unlabeled pool are used when constructing $\mathcal{S}_t$.

Outcome-Validated Credit Assignment

After $\mathcal{B}_t$ is labeled, the segmentation model is retrained once:

$$M_{t+1} = f(\mathcal{L} \cup \mathcal{B}_t).$$

For each reward region $q \in \mathcal{R}$, we measure the improvement:

$$\Delta_t(q) = \ell(M_t, q) - \ell(M_{t+1}, q),$$

where ℓ is the region-level segmentation loss or pixel error. Positive $\Delta_t(q)$ indicates that retraining improved the model on that reward region.

Each selected region $r \in \mathcal{B}_t$ receives credit by matching it to reward-set regions whose error decreased:

$$G_t(r) = \sum_{q \in \mathcal{R}} w_t(r, q) \Delta_t(q), \quad (4)$$

where

$$w_t(r, q) = \frac{\exp(\text{sim}(z_t(r), z_t(q))/\tau_c)}{\sum_{q' \in \mathcal{R}} \exp(\text{sim}(z_t(r), z_t(q'))/\tau_c)}. \quad (5)$$

This is an outcome-conditioned credit signal rather than a causal-counterfactual estimate. It provides a scalable preference signal from one retraining per round.

The final observed utility is

$$u_t(r) = G_t(r) + \alpha \Phi_t(r) - \gamma D_t(r), \quad (6)$$

where $\Phi_t(r)$ measures failure-mode match and $D_t(r)$ penalizes redundancy with already selected regions and high-utility regions in E_t . All terms are normalized within each round before preference pairs are formed.

Dataset	Images	Categories/classes	Split policy
MVTec AD	5,354	15 categories	50/15/15/20 stratified
VisA	10,821	12 categories	50/15/15/20 stratified
CT-Defect	13,562	inclusion/void/crack	official train/test retained

Table 5: Dataset split policy. Public benchmark results are comparisons within the supervised active-learning protocol, not standard unsupervised anomaly-detection leaderboard results.

Item	Value
Image size	512×512
Region size / stride	128×128 / 64
Max regions per image	49
Initial labeled seed $\mathcal{L}_0$	5%
Acquisition rounds T	9
Per-round query budget b	5%
Total query budget B	45%
Final labeled fraction	50%
Shortlist size	$K = 5b$
Subtle threshold τ	1% image area
Seeds	5 matched active-learning seeds
Preference pairs per round	at most 32
Max preference pairs total	at most 288

Table 6: Active-learning protocol used in all experiments.

Item	Value
Segmentation model	SegFormer-B2
Encoder initialization	ImageNet-pretrained MiT-B2
Optimizer	AdamW
Learning rate	6×10^{-5}
Weight decay	10^{-2}
Batch size	8
Augmentations	standard geometric augmentations
Loss masking	labeled regions only

Table 7: Segmentation training configuration.

Planner Objective

The planner solves a redundancy-aware subset problem over the candidate shortlist $\mathcal{C}_t$:

$$\mathcal{B}_t = \arg \max_{\mathcal{B} \subseteq \mathcal{C}_t, |\mathcal{B}|=b} \left[\sum_{r \in \mathcal{B}} \hat{v}_t(r) + \eta \Gamma_t(\mathcal{B}) - \lambda \Omega(\mathcal{B}) \right]. \quad (7)$$

Redundancy is computed as

$$\Omega(\mathcal{B}) = \sum_{\substack{r_i, r_j \in \mathcal{B} \\ i < j}} [\omega \cos(z_t(r_i), z_t(r_j)) + (1 - \omega) \text{IoU}(r_i, r_j)]. \quad (8)$$

Failure coverage is

$$\Gamma_t(\mathcal{B}) = 1 - \prod_{r \in \mathcal{B}} (1 - \text{sim}(z_t(r), F_t)). \quad (9)$$

The objective is optimized greedily: starting from directly accepted regions, the solver repeatedly adds the candidate with largest marginal gain until the budget is filled.

Item	Value
Fast policy	Qwen2.5-VL-3B-Instruct
Planner	Qwen2.5-VL-7B-Instruct, frozen
Preference objective	Bradley–Terry with DPO-style reward
LoRA rank	16
LoRA scaling	32
LoRA dropout	0.05
Policy learning rate	5×10^{-5}
Policy update	one epoch per round
Preference temperature β	0.1
Escalation margin δ	0.05
Memory similarity threshold	0.60
Experience memory size	256 regions
Failure clustering	k -means, $k = 5$
Planner weights	$\eta = 0.15, \lambda = 0.25, \omega = 0.5$

Table 8: Fast-policy and planner hyperparameters.

Preference Optimization

For a validated pair $r_i \succ r_j$, the generated response y_{r_i} is treated as preferred and y_{r_j} as rejected. Because r_i and r_j are distinct candidate contexts, we use a Bradley–Terry pairwise ranking objective parameterized by a DPO-style implicit reward:

$$\rho_\theta(r) = \log \frac{\pi_\theta(y_r | c_r, \mathcal{S}_t)}{\pi_{\text{ref}}(y_r | c_r, \mathcal{S}_t)}. \quad (10)$$

The policy is optimized with

$$\mathcal{L}_{\text{pref}}(\theta) = -\mathbb{E}_{(r_i \succ r_j)} [\log \sigma(\beta(\rho_\theta(r_i) - \rho_\theta(r_j)))]. \quad (11)$$

Per-Category Results

Per-category results test whether SAGE’s gains are concentrated in only a few categories or are consistent across object and texture types. We report final Dice at the last active-learning round for MVTec AD and VisA. Gains vary across categories: SAGE helps most on low-contrast texture-like defects and less when defects are visually salient enough for uncertainty or region-selection baselines to surface early.

Additional Ablations

Failure Cases and Low-Gain Settings

SAGE does not improve every category or round equally. Its smallest gains occur in categories where defects are large, high-contrast, or already surfaced early by uncertainty-based acquisition. In such cases, the benefit of model-state-conditioned policy evolution is reduced because static baselines already query useful regions.

MVTec category	Random	Entropy	CoreSet	RALIS	SAGE	Δ vs RALIS
bottle	64.2	70.5	69.8	75.6	78.2	+2.6
cable	56.8	63.7	62.4	69.1	73.8	+4.7
capsule	55.1	62.9	61.8	68.4	71.1	+2.7
carpet	60.7	67.6	66.9	72.5	78.6	+6.1
grid	58.4	65.1	64.2	70.7	73.4	+2.7
hazelnut	63.8	69.4	68.9	74.0	76.6	+2.6
leather	61.5	68.8	67.9	73.4	79.8	+6.4
metal nut	59.3	66.2	65.4	71.3	74.6	+3.3
pill	54.7	61.8	60.5	67.2	71.6	+4.4
screw	52.9	60.7	59.9	66.0	69.3	+3.3
tile	62.1	68.5	67.6	73.0	78.1	+5.1
toothbrush	60.4	66.9	66.1	72.1	74.0	+1.9
transistor	57.2	64.4	63.6	69.8	72.9	+3.1
wood	63.5	69.2	68.1	73.6	77.8	+4.2
zipper	56.1	63.1	62.5	68.9	73.9	+5.0

Table 9: Per-category final Dice on MVTec AD. Values are averaged over five matched active-learning seeds. Gains are not uniform: SAGE helps most on subtle texture-like categories and less where baseline uncertainty already surfaces salient defects.

VisA category	Random	Entropy	CoreSet	RALIS	SAGE	Δ vs RALIS
candle	52.0	60.2	59.1	67.3	69.7	+2.4
capsules	50.8	58.9	57.7	65.8	69.8	+4.0
cashew	55.6	62.7	61.8	69.0	71.7	+2.7
chewinggum	56.4	63.5	62.8	70.2	75.4	+5.2
fryum	53.3	61.1	60.2	67.8	72.7	+4.9
macaroni1	51.7	59.5	58.8	66.4	69.5	+3.1
macaroni2	52.4	60.6	59.7	67.1	72.9	+5.8
pcb1	54.1	61.9	60.8	68.6	71.9	+3.3
pcb2	53.6	61.2	60.1	68.1	70.4	+2.3
pcb3	55.2	62.8	61.4	69.7	75.6	+5.9
pcb4	54.8	62.1	61.0	69.1	73.0	+3.9
pipe fryum	50.1	58.4	57.0	64.8	68.9	+4.1

Table 10: Per-category final Dice on VisA. Values are averaged over five matched active-learning seeds.

A second low-gain setting occurs when the reward set contains few close analogues of the current failure mode. The credit-assignment signal transfers reward-set improvement to selected regions by descriptor similarity; when matches are weak, the resulting preference pairs become noisier. This explains occasional non-monotonic learning-curve behavior and categories where SAGE remains competitive but improves only modestly over RALIS.

Dataset	Spearman ρ	Kendall τ
MVTec AD	0.72 ± 0.04	0.55 ± 0.05
VisA	0.68 ± 0.05	0.50 ± 0.04
CT-Defect	0.70 ± 0.04	0.53 ± 0.05

Table 11: Agreement between the single-retraining credit score and an expensive leave-one-out approximation.

These cases clarify the intended operating regime: SAGE is most useful when targets are small, low-contrast, confidently missed, and when the annotation batch must balance failure coverage against redundancy.

Reward-Set Sensitivity

Because SAGE uses a labeled reward set to construct outcome feedback, we evaluate how much calibration data is required. The active query budget is fixed while only the reward-set size varies. The total labeled fraction counts the final active training labels, validation labels, and reward-set calibration labels.

Credit-Assignment Validation

To check that the single-retraining credit signal is meaningful rather than noise, we compare it against an expensive leave-one-out approximation that re-trains with each selected region removed. The two scores agree strongly (Table 11), supporting the use of the cheaper outcome-conditioned signal.

Component Ablation

Table 14 reports the full component ablation, which extends the main-paper ablation with the *w/o experience memory* and *uncertainty-only state* variants. Removing the experience memory or reducing the state to uncertainty alone lowers both overall and subtle-region Dice, confirming that the

Method	MVTec AD				VisA			
	R@50	R@100	Reg.-90↓	Area Err.↓	R@50	R@100	Reg.-90↓	Area Err.↓
Random	22.4	36.2	271	24.9	17.8	29.6	424	27.5
Entropy	39.1	57.6	196	19.8	32.7	50.4	322	21.7
BALD	41.3	60.4	184	18.7	35.1	53.8	300	20.3
CoreSet	35.8	54.9	205	17.9	31.4	49.1	336	19.1
VAAL	43.6	62.7	176	16.6	37.6	56.9	284	18.2
EquAL	45.2	64.1	168	16.1	39.8	58.4	274	17.6
RALIS	49.8	68.7	151	14.4	45.9	65.1	238	15.5
SAGE	58.7	77.4	119	10.4	55.1	75.6	194	11.6

Table 12: Offline review-prioritization simulation on the public benchmarks. R@K is target-region recall in the top- K reviewed regions; Reg.-90 is regions reviewed to reach 90% recall; Area Err. is defect-area error.

Reward set	Total labeled	Dice	AULC	Subtle Dice
2%	67%	72.1 ± 1.2	63.7 ± 1.3	53.8 ± 1.6
5%	70%	73.8 ± 1.0	65.1 ± 1.1	57.2 ± 1.4
10%	75%	74.6 ± 0.9	66.0 ± 1.0	59.1 ± 1.2
15%	80%	75.0 ± 0.8	66.6 ± 0.9	60.7 ± 1.1

Table 13: Reward-set sensitivity averaged over MVTec AD and VisA.

improved even when another region in the batch caused most of the improvement. We therefore validate the proxy against sampled leave-one-out retraining, but the approximation is not exact.

Finally, the review-prioritization evidence is retrospective. The public benchmark simulation measures whether confirmed target regions are ranked earlier; it does not measure expert review time in a live setting. A prospective study with domain experts is left to future work.

richer round-wise state contributes beyond uncertainty.

Variant	Dice	AULC	Subtle Dice	Cost↓
SAGE	75.0 ± 0.8	66.6 ± 0.9	60.7 ± 1.1	0.54
w/o planner	72.4 ± 1.1	63.9 ± 1.2	55.0 ± 1.4	0.29
Planner-All	75.5 ± 0.8	66.9 ± 0.9	61.3 ± 1.1	1.00
Frozen VLM	72.9 ± 1.1	64.1 ± 1.2	54.8 ± 1.5	0.72
w/o experience memory	73.6 ± 1.0	64.9 ± 1.1	56.9 ± 1.4	0.63
Uncertainty-only state	71.8 ± 1.2	62.8 ± 1.3	49.8 ± 1.7	0.61
Image-level acquisition	69.5 ± 1.4	60.7 ± 1.5	45.9 ± 1.9	0.48

Table 14: Component ablation averaged over MVTec AD and VisA.

Offline Review-Prioritization Simulation

SAGE is intended as a review-prioritization layer rather than an autonomous decision system. The offline simulation ranks candidate regions as if they were presented to an expert reviewer and then uses ground-truth masks to measure how quickly confirmed target-positive regions are surfaced.

Limitations

SAGE uses a labeled reward set to construct outcome feedback. We treat this as a one-time calibration resource rather than part of the active query budget, but this assumption depends on the availability of archived labeled examples. The reward-set sensitivity study quantifies how performance changes as this calibration set is reduced.

The credit-assignment procedure is outcome-conditioned rather than causal-counterfactual. It can assign high utility to a selected region that resembles reward regions whose losses